Adrit Panday

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EDUCATION

Honors Bachelor of Science, Computer Science Specialist (Co-op) Expected Graduation: October 2027
University of Toronto Scarborough - Toronto, ON

- Minor in Astrophysics and Astronomy
- Relevant Coursework: Operating Systems, Artificial Intelligence, Data Structures & Algorithms, Systems Programming

WORK EXPERIENCE

ML Software Developer, Sunnybrook Hospital || Python October 2024 - May 2025

- Engineered **automated** Python preprocessing **pipelines** for histopathology **tumor** imaging, reducing manual data preparation time by **64%** and enabling faster and more scalable model training workflows.
- Developed a multi-format **image parsing** framework supporting .svs and .tif whole-slide scans with QuPath and OME-XML integration, expanding **dataset compatibility** across multiple research pipelines by **40%**.
- Optimized **deep learning** training pipelines by implementing custom preprocessing classes for flow recomputation, geometric augmentation, and resolution normalization using **PyTorch**, improving training and evaluation efficiency by **35%**.
- Enhanced AI-assisted **tumor segmentation** workflows within Paracell's **research** platform while **optimizing GPU**-based preprocessing throughput and memory usage to **support large-scale** model training on high-resolution whole-slide images.

PROJECTS

Robot Soccer Autonomous System || C/C++, Linux, OpenCV/EV3 September 2024 - December 2024

- Developed **autonomous robot** control software integrating webcam perception, blob tracking, and Bluetooth actuation on LEGO EV3 hardware, helping team place **Top 8** in RoboSoccer competition.
- Implemented **finite-state decision logic** for ball interception, goal pursuit, and opponent avoidance with sub-second action switching during dynamic gameplay.
- Built calibration pipeline for perspective correction, heading estimation, and object localization across a **170 cm × 115 cm** playfield under noisy vision input.
- Tuned motion commands and control timing through iterative live-match testing, enabling smoother turns and target pursuit.

Probabilistic Robot Localization || C/C++, Linux September 2024 - December 2024

- Implemented histogram / Markov localization maintaining belief states across **grid intersections** × **4 orientations** under uncertain starting position and noisy color sensor readings.
- Designed probabilistic belief updates using sensor observations and motion transitions to continuously refine robot positioning.
- Improved **navigation accuracy** through filtering, repeated measurements, and robust recovery from ambiguous states.

Neural Network Training Engine (From Scratch) || C, Linux February 2026 - March 2026

- Engineered 1-layer and 2-layer neural networks from scratch in C, implementing forward propagation, full error **backpropagation**, and explicit weight update mechanisms without external ML libraries.
- Derived and implemented full error backpropagation across hidden layers, carefully managing gradient flow and weight update sequencing to prevent update leakage and ensure stable convergence.
- Trained models on MNIST and CIFAR-10 using structured experimental workflows, achieving high-**80% classification accuracy** on MNIST, and analyzing performance gaps between shallow and multi-layer architectures.
- Experimented with hidden layer sizes, activation functions, and input scaling to mitigate saturation and optimize convergence behavior during training.

GLOW || Node.js, Express, MongoDB, Next.js, Jira May 2025 - August 2025

- Built a **full-stack** platform for **real-time** nearshore water temperature analysis across the Great Lakes, including backend APIs, data storage, authentication, and interactive frontend components.
- Containerized application services using **Docker** and implemented CI/CD **automation** via GitHub Actions, ensuring reproducible builds, automated testing covering ~**90%**, and production deployment readiness.
- Designed scalable RESTful APIs and MongoDB data models supporting secure ingestion, retrieval, and manipulation of high-volume geospatial environmental data with JWT-based authentication.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, Java, SQL
Robotics & Vision: OpenCV, Sensor Fusion, Localization, State Machines, Motion Planning, Real-Time Control
Machine Learning & AI: PyTorch, TensorFlow, NumPy, pandas, Scikit-learn, LLM APIs
Frameworks: Node.js, Express, Next.js
Infrastructure & Tools: Git, GitHub Actions, Docker, Linux (Ubuntu), Jira, MongoDB, VS Code